

Cloud Computing Lab File



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Subject Name: Cloud Computing Lab

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Lab 2: To Install and configure VMware Workstation Pro / Oracle Virtual Box for creating the Virtual Machines, cloning and deleting VMs.

Aim: Exploring VMWare Workstation Pro/oracle virtual box to create the virtual Machines

Perform the following:

1. Making Clone in Virtual Box

Choose a Virtualization Platform

Local options: VMware Workstation, VirtualBox, Hyper-V.

Install the Virtualization Software

Download and install the software on your system.

Create the First Virtual Machine

- i) Start the VM Creation Wizard:
- ii) Open your virtualisation platform.
- iii) Select Create New Virtual Machine (or similar option).
- iv) Choose an Operating System:
- v) Select an OS (Linux, Windows, etc.) for your VM. You may

need an installation ISO or prebuilt image.

vi) Allocate Resources:

vii) Assign resources such as:

viii) CPU: Number of cores.

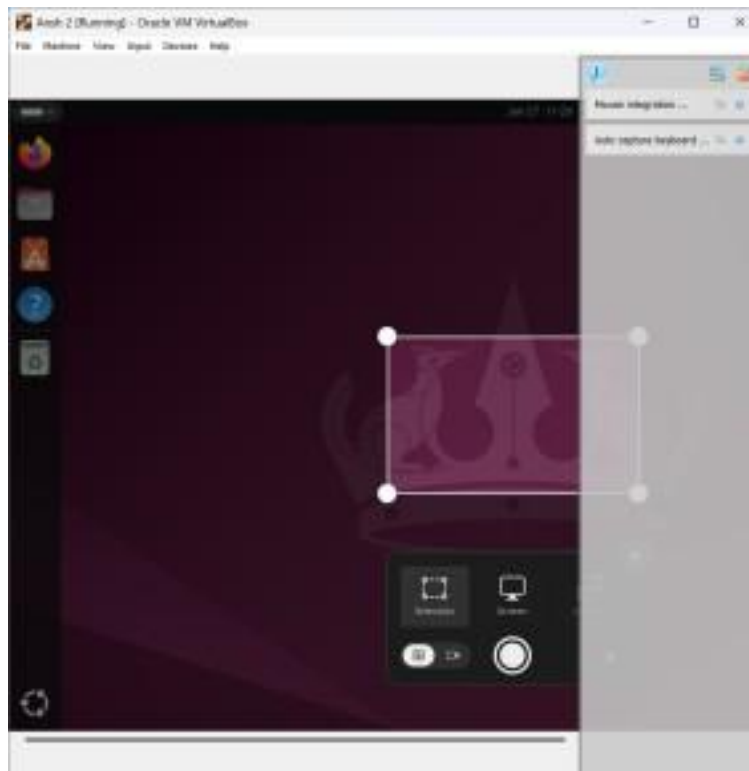
ix) RAM: Memory size (e.g., 2 GB, 4 GB).

x) Disk Storage: Storage size (e.g., 20 GB).

xi) Configure Network

xii) Select a network type (NAT, Bridged, or Host-only). xiii) Finish Setup

xiv) Complete the configuration and start the VM installation.



2. Install two or more Guest Operating Systems on all the VMs.

1. Prepare Installation Media

Download ISO files or installation images for the guest OS (e.g., Windows, Linux distros like Ubuntu, CentOS).

Place the files in an accessible folder on your system.

2. Open the Virtualization Software

Launch your virtualization platform (e.g., VMware Workstation, VirtualBox, Hyper

3. Start VM Creation for the First OS

Create a New VM:

Click New Virtual Machine (or equivalent). Select

Installation Source:

Attach the ISO file for the first guest OS during setup. Assign

Resources:

Allocate CPU, RAM, and storage as needed. Start the

VM:

Boot the VM to launch the guest OS installer.

4. Install the First Guest OS

Follow the guest OS installation process: Select language, time zone, etc.

Partition the disk (if required). Install the OS files.

Reboot the VM once the installation completes.

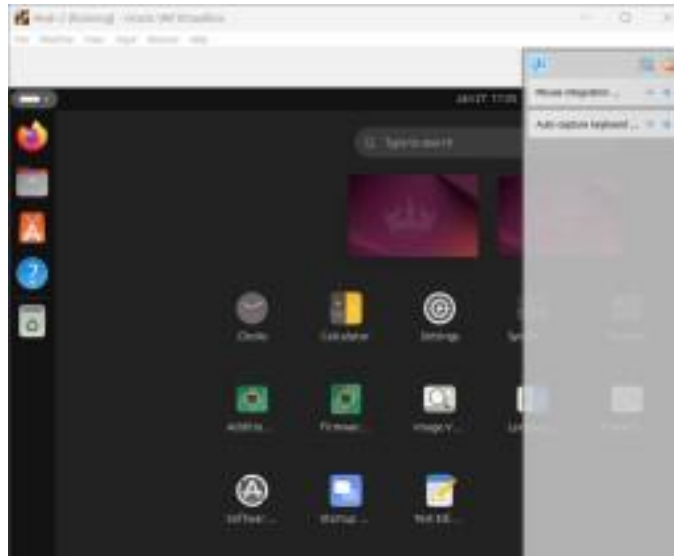
Set up a username, password, and additional settings.

3. Run simple applications or programs on all the VMs.

Start All Virtual Machines

Open your virtualization software (e.g., VMware Workstation, VirtualBox, or Hyper V).

Power on each VM and log in to the guest operating systems.



4. Cloning of Virtual Machine: Full Cloning, Linked Cloning

1. Full Cloning

- Creates a completely independent copy of the VM.
- Power Off the Base VM:
 - Shut down the VM you want to clone to ensure consistency.
- Open Virtualisation Software:
 - Open VMware Workstation, VirtualBox, or any virtualisation platform you're using.
- Right-Click the VM:
 - Select the VM you want to clone and right-click on it.

Choose Clone Option:

In VMware Workstation:

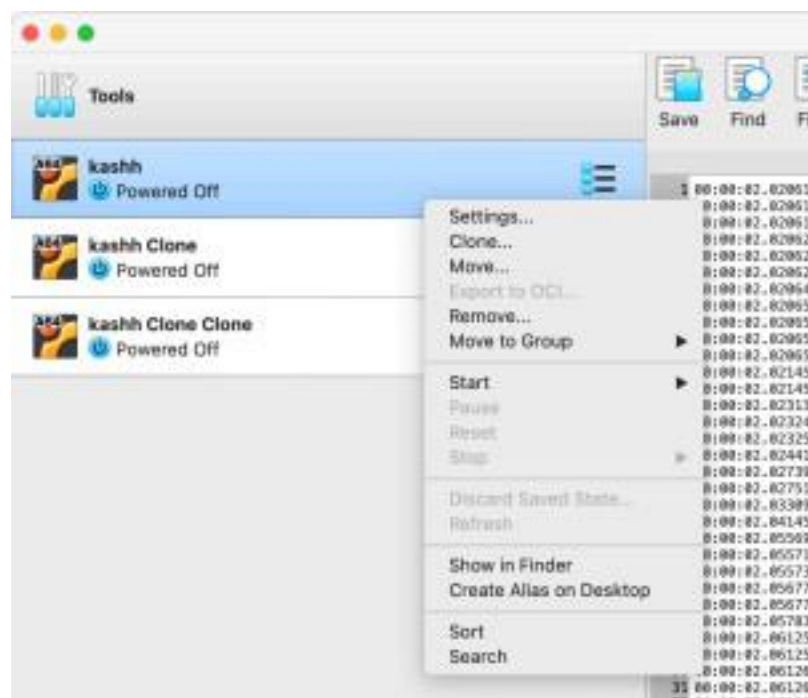
Go to Manage > Clone. In

VirtualBox:

Select Clone from the context menu. Select

Cloning Method:

Choose Full Clone when prompted.



- Creates a lightweight copy that depends on the base VM.
- Power Off the Base VM or Take a Snapshot:
- Shut down the base VM or take a snapshot to ensure a clean state.
- Open Virtualization Software:

- Open VMware Workstation or VirtualBox.
- Right-Click the VM:
- Select the VM you want to clone and right-click.



5. Add a USB Controller to Virtual Machine. Power Off the VM:

Ensure the VM is turned off before making hardware changes. Open VMware Workstation:

Launch VMware Workstation or Player and select the VM you want to modify. Edit VM Settings:

Right-click on the VM name and select Settings (or click Edit virtual machine settings).

Add USB Controller:

In the Virtual Machine Settings window, click on the Add button at the bottom. Select USB Controller from the list of available devices and click Next.

3. Closing Virtual Machines & Exiting Workstation Pro

a. Hard and Soft Power Off

b. Configure Virtual Machine to Run in the Background

c. Pause and Unpause a Virtual Machine

d. Pause all Virtual Machine without interacting Workstation Pro / Oracle

Hard and Soft Power Off

Hard Power Off: In VMware, go to VM > Power > Power Off; in VirtualBox, select Power Off in the Close menu.

Soft Power Off:

In VMware, go to VM > Power > Shut Down Guest; in VirtualBox, select Send the shutdown signal.

b. Configure Virtual Machine to Run in the Background VMware:

Start VM, minimise VMware

Workstation to run in the background. VirtualBox:

Start VM, then choose Machine > Close > Save

the machine state.

c. Pause and Unpause a Virtual Machine

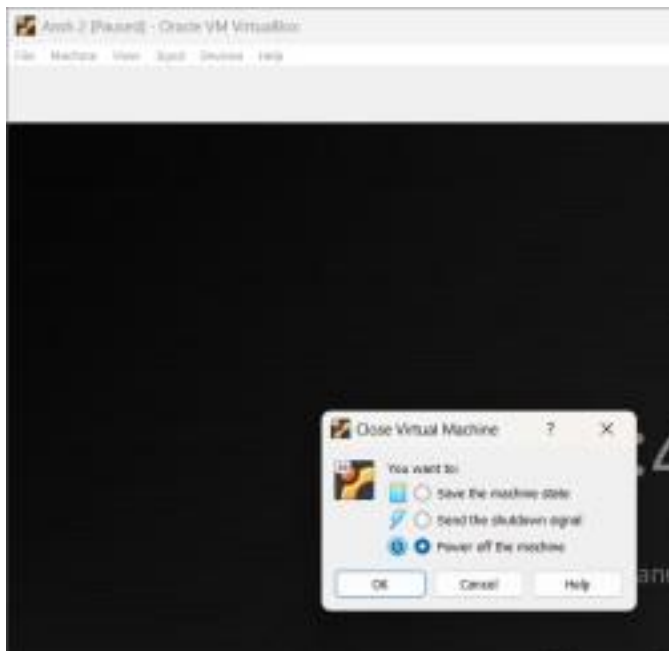
VMware: Go to VM > Pause to pause; VM > Unpause to resume.

VirtualBox: Go to Machine > Pause to pause; Machine > Resume to unpause.

d. Pause All Virtual Machines Without Interacting with Workstation Pro / Oracle VM VirtualBox

> Pause All to pause all VMs.

VirtualBox: Right-click the VM list and select Pause All, or use VBoxManage command to pause VMs.



4. Delete a virtual machine from the disk.

Open Oracle VirtualBox: Launch VirtualBox and go to the VirtualBox Manager. Select the VM: Click on the VM you want to delete from the list. Right-click and Choose "Remove":

Right-click the selected VM.

Choose Remove from the context menu.

